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Inheritance of resistance in maize to the virus of streak disease in East Africa

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SUMMARY

Maize plants selected from lines raised in South Africa, and resistant to an East African isolate of streak virus, gave rise on selfing to lines that bred true for resistance through four generations.

Heterozygotes from crossing resistant by susceptible lines reacted to infection by inoculation with the vector in a manner intermediate between the parents; neither allelomorph was fully dominant. The F_2 generation from the cross segregated in the expected 1:2:1 ratio. Backcrosses segregated 1:1.

The evidence is interpreted as showing control of resistance by a major gene, but with modification in details of the reaction attributed to minor genes.

INTRODUCTION

Although the virus causing streak disease is prevalent in maize, *Zea mays* L., in many parts of Africa, no substantial resistance appears to have evolved among varieties grown in this continent. A notable exception is the line known as 'Peruvian Yellow' (but believed to be an early importation into South Africa from Argentina (Gorter, *in litt.*)). A less marked resistance was also reported in a Barberton race of 'Hickory King'. Resistance in these was recognized and exploited by the Cotton Research Station at Barberton (*Empire Cotton Growing Corporation, Progress Reports* from 1931 to 1944). After 1944 the material was transferred to the government of South Africa and further developed by Gorter (1953, 1959), who found that hybrids of Peruvian × Hickory tended to tolerate the virus although showing no marked resistance to infection. Derivatives of these gave satisfactory control in areas of high natural streak infection in South Africa (Gorter, 1959). Field experiments had given no guidance to the manner in which resistance was inherited, which appeared not to conform with Mendelian ratios.

Our purpose has been to find whether a genetic hypothesis that would assist the breeding of streak-resistant maize would emerge from the more exact studies possible in glasshouses.

EXPERIMENTAL METHODS AND MATERIALS

Plants. Maize seedlings were raised in gauze-protected glasshouses, in a fertile compost, usually formed into soil-blocks. Selected plants were transplanted (at 30–40 days from sowing) to 4-gallon containers, in which they were grown to maturity

in a large glasshouse. At all stages the plants received a dilute inorganic fertilizer; they were a good colour and growth was normally strong. Insect infestations during the second stage were controlled with a systemic insecticide added to the soil.

The usual methods for controlling pollination were applied.

Virus. McClean (1947) showed that there exist several forms—supposedly strains—of the streak virus. We needed the most virulent field strain present in East Africa. As a first attempt to isolate different strains from local sources we fed the vector on the earliest chlorotic spots appearing on infected maize seedlings; but on further transfer four such isolations gave severe symptoms that were indistinguishable. One of these isolates was used in all the experiments here reported.

In further search for virus strains, samples of severely streak-diseased leaves were collected from ten localities in Kenya, Uganda and Tanzania; but transfers from these to susceptible maize gave no recognizable differences in symptoms. One isolate (from Arusha, Tanzania) appeared initially to be more virulent to resistant lines; however, this character was lost in later passages through maize.

The foregoing tests did not indicate that more than one virus was present and we regard this as McClean's no. A (1947).

Virus inoculation. Since no method is known for successfully inoculating this virus mechanically into a plant, it was necessary to rely on an insect vector; and we used *Cicadulina mbila* (Naude) China from bulk cultures of an *active* race (Storey, 1932). Normally each individual insect of this homozygous line reliably transmits the virus, but their efficiency varies; and sometimes a prolonged test feed may be necessary to ensure infection.

Our standard method of testing maize seedlings was to confine two adult insects together in a glass-tube leaf-cage (as illustrated in Storey, 1928, p. 2), following an acquisition feed on a streaked leaf for at least 4 days. A cage was clipped on to the first or second leaf of each seedling about 10 days from sowing and removed after 2 days. If no symptoms had appeared on the seventh day, a second independent but similar test feed was given.

To test the reliability of this procedure, eight replications of 100 plants of a susceptible line were exposed for 2 days to single insects. Following the independence hypothesis, enunciated by Watson (1936) for aphids and confirmed by Storey (1938) for *Cicadulina mbila*, if q is the probability that a single insect will fail, then q^2 is the probability that two insects together will fail, and q^4 the probability that two successive two-insect tests will fail. Our results revealed minor anomalies on statistical analysis; but the mean value for q was 0·1757 and of q^4 0·0009, and the probability of failure was therefore low.

The above hypothesis is valid only if the test plants are uniformly susceptible, as was emphasized by Watson (1936). We believe that the susceptible maize line used above is so, for a confirmatory test by our standard method gave all of 106 seedlings diseased in eleven separate experiments. Deviations, as encountered in the studies described later, can fairly be attributed to qualities possessed by the plants and not to any failure in inoculation.

THE DEVELOPMENT OF STREAK DISEASE IN MAIZE

The way symptoms develop in an inoculated susceptible plant has often been described (e.g. McClean, 1947).The virus produces no visible effects at the point of inoculation but passes rapidly downwards into the young tissue at the leaf bases (Storey, 1928). The first visible symptoms are small discrete chlorotic areas, circular or elongated parallel to the veins, appearing often as early as 3–4 days after inoculation. Individual spots do not enlarge as the leaf extends, but they may be larger and more frequent in later-formed areas of leaves. Glasshouse seedlings inoculated through the first or second leaf normally show a nearly uniform pattern of chlorotic and green areas over the whole of the sixth to eighth leaf (fig. 2 of Storey, 1925). Later-formed leaves may show a less severe chlorotic pattern.

In resistant maize, divergences may occur in the size of the chlorotic areas, in the frequency of their distribution and in the persistence with which they appear in later-formed leaves. It is on such differences that the classification of infection types is based in the next section.

CLASSIFICATION OF INFECTION TYPES IN EXPERIMENTAL PLANTS

Initially the following eight classes were used in recording glasshouse experiments (Pl., figs 1–6).

O: no symptoms at any stage.

A– –: one or a few small circular chlorotic spots, confined to a single leaf without further development.

A–: as A– – but spots developing on several leaves; none on the later leaves (fig. 1).

A+: spots more numerous than in A– and tending to be elongated slightly. May continue rarely to appear in small numbers on all leaves to maturity of the plant (fig. 2).

B–: spots larger, mostly elongated and more numerous than in A+ and distribution maintained normally to maturity (fig. 3).

B+: as B–, but generally more severe; differentiation between these two classes often uncertain (fig. 4).

C– and C+: most severe form of symptoms, chlorotic areas occupying some half or more of the leaf surface (figs. 5, 6). Dividing line between these two classes often uncertain: except that sometimes in C+ in seedlings early leaves may develop deformed and stunted. This character has been regarded as placing the plant in C+, but its absence did not certainly place it in C–.

The earlier classes constitute the form of resistance we have studied. However, a nearly continuous range of phenotypes can occur; and allocation between adjacent classes has often been uncertain. A numerical expression of 'severity', by a systematized process of scanning leaves by microscope and so measuring the proportion of the total leaf area occupied by chlorotic spots and streaks, proved impracticable in a plant breeding programme, and we have therefore relied on eye-assessment of classes. This

was sometimes difficult and unreliable; all classifications were decided by one or both of the authors on visible symptoms 20–30 days after inoculation.

EXPERIMENTAL RESULTS

The basic resistant material of these studies was three Peruvian × Hickory lines selected from eight provided by Dr Gorter, his nos. B.P.P. G. 35, G. 43, G. 45 (our accession numbers, SR. 255, 257, 259). No. AFRO 54300, a derivative of Colombia 2, maintained by us for some 12 years of sibbing (but with only limited selection for reaction to the streak virus), was used as a susceptible line in our crossing experiments.

Table 1. *Combined distributions of families from SR.* 255, 257 *and* 259 *and inbred derivatives by selfing class O and A– – phenotypes*

Generation	No. of samples of families	Total plants tested	30-day distribution (% of totals)						
			O	A– –	A–	A+	B– & B+	C–	C+
P	3	92	16·3	28·3	26·1	7·6	18·5	3·3	0
Total of S_1 to S_4	81	1924	18·9	30·7	37·7	8·9	3·6	0·2	0

Inbreeding resistant phenotypes

Initial tests of the three inbred lines as provided by Dr Gorter showed a distribution, according to our scale of classification, as in the first line of Table 1. In the total of ninety-two plants tested, 78% fell in the O and A classes, the remainder in B and C classes (but none in C+). The second line gives the summation of the distributions in four generations of successive selfs of class O and A– –: eighty-one families, including in all 1924 plants. There was a narrowing of the distribution towards the left-hand side of the table: 96% of the total in the O to A classes, with a small overflow into B and C–.

Statistical examination revealed considerable heterogeneity in the distribution of classes O and A in generation and family totals. SR. 255 gave more consistent data than SR. 257 and 259, although the behaviour of these was generally similar. For this reason, and also because SR. 255 entered more fully into later experiments in crossing, only the data from inbreeding within this line are summarized in Table 2.

SR. 255 progenies from class O parents were aggregated mainly in the classes O, A– – and A–: distributions between these three showed no significant heterogeneity in generation totals, although, within each, constituent families tended to be heterogeneous. There was in some a small overflow into A+ and B classes, especially in the S1.

Distributions in families deriving from A– – and A– parents are summarized as combined totals of all S generations in the later part of Table 2. They differ from O in showing a shift towards the right-hand side of the table; and in families grouped as (*b*) this extended considerably into A+ and B. Thus, A– – gave sixteen families resembling O in being confined mainly to O and A classes, but also six families extending into B. From A– parents this tendency was more evident, three families being in group (*a*) and six in (*b*).

Table 2. *SR. 255: distributions of families inbred from classes O, A– – and A– parents*

Class of parents	Generations	Groups	No. of families	Total plants tested	30-day distributions (% of totals) O	A– –	A–	A+	B–, B+	C–, C+
O	S_1	—	3	68	23·5	36·8	35·3	1·5	2·9	0
	S_2	—	4	39	28·2	38·5	33·3	0	0	0
	S_3	—	3	41	31·7	34·1	29·3	4·9	0	0
	S_4	—	2	76	27·6	38·2	34·2	0	0	0
A– –	S_1–S_4	(*a*)	16	375	12·3	42·9	42·1	2·7	0	0
	S_2–S_4	(*b*)	6	170	11·2	10·6	37·1	20·0	21·2	0
A–		(*a*)	3	96	10·4	37·5	49·0	3·1	0	0
		(*b*)	6	236	8·1	7·2	35·2	19·9	29·2	0·4

Table 3. *SR. 255: distributions of crosses of resistant by susceptible phenotypes*

Generations	Classes of parents	Groups	No. of families	No. of plants	30-day distributions (% of totals) O to A–	A+	B–, B+	C–	C+
F_1	O × C	(*a*)	5	69	0	0	81·2	18·8	0
	A– – × C	(*b*)	12	194	0	0·5	68·6	30·4	0·5
F_2	B selfed	(*a*)	15	479	20·5	5·8	32·4	17·9	23·4
	C– selfed	(*b*)	10	232	14·3	10·8	20·6	31·4	22·9
F_3	O selfed	(*a*)	3	66	59·1	21·2	19·7	0	0
		(*b*)	1	23	13·0	4·3	16·1	26·1	30·4
	A– – & A– selfed	—	7	132	68·2	18·9	12·9	0	0
	A+ selfed	(*a*)	1	21	33·3	52·4	14·3	0	0
		(*b*)	1	23	13·0	17·4	13·0	26·1	30·4
	B–, B+ selfed	—	8	181	19·3	3·9	38·1	18·8	19·9
	C– selfed	(*a*)	2	48	6·2	10·4	16·7	47·9	18·8
		(*b*)	1	24	0	0	0	0	100
	C+ selfed	—	4	95	0	0	0	0	100

Selective inbreeding can clearly maintain lines that react to the streak virus to give phenotypes mainly in the O and A classes. These classes constitute a resistant category; all are equally effective and scarcely distinguishable in the field. Nevertheless, some inbred families from resistant parents may be distributed into B and even into C−. We have considered non-inherited factors that might have caused this diversity. One is environment; and an experiment, repeated at 2-monthly intervals over 1 year, examined mean maximum temperatures during the incubation period of the virus in a pure resistant line. Reactions at the highest (39° C. in January) were confined to the O to A− classes; at the lowest (28° C. in September) there was considerable extension into B classes. The evidence suggests a partial inactivation of the streak virus at higher temperatures; studies with more accurate control equipment would be necessary to substantiate this suggestion.

A second possible factor was studied by exposing seedlings to infection by groups of one, two, four and eight vectors; this caused a shift to the right in family distributions, as indeed might be expected on the assumption that the probability of including insects of high efficiency rises as the number in each group is increased (Storey, 1938).

Despite this limited evidence, the data recorded in Table 2, deriving from many experiments replicated at all seasons of the year with randomized samples of the vector, rather suggest that the diversity is due to inherited modifying factors.

Crossing resistant by susceptible phenotypes

Resistant parents were class O and A− − phenotypes, selected from families of narrow distribution in the S_1 and S_2 generations from inbreeding SR. 255 (Table 2). Susceptibles were derived from AFRO 54300, which on selfing had given distributions confined mainly to C− and C+: twelve families totalling 325 plants, of which 96·3% were classified C+ and the remainder C− (2·5%) and B (1·2%).

The data from three generations of crosses are summarized in Table 3. The F_1 generation totals are grouped according to the classes of the resistant parents, (*a*) class O and (*b*) A− −. In both groups family distributions were confined to the B and C− classes with very minor overflows into A+ and C+ in (*b*).

Table 4. *Distributions of backcrossed families from F_1 of Table 3*

Class of parent from F_1	No. of families	Total plants	30-day distributions (% of totals)			
			O to A−	A+	B & C−	C+
1. Recurrent parent susceptible						
B	5	114	0·9	0	53·5	45·6
C−	2	82	3·7	0	56·1	40·2
2. Recurrent parent resistant						
B	3	71	42·2	5·6	50·7	1·4
C−	9	210	44·3	10·5	41·9	3·3

Selfing of F_1 phenotypes, classes B and C−, gave the data of (*a*) and (*b*) in the F_2 of the table. The total family distributions, grouped as resistant, O to A+, intermediate, B and C−, and susceptible, C+, showed a good agreement with a 1:2:1 ratio, in (*a*) and (*b*) separately and in the totals of each.

The F_3 generation was tested mainly to study the validity of the grouping in the preceding paragraph. In general this validity was confirmed. Thus O, A− − and A− selected parents gave family distributions conforming with the assumption that they were homozygous resistants (except one family, in F_3 (b), explicable as an error in selection). On the other hand, A+ is not reliably homozygous; of two tested, one was homozygous and one heterozygous. Again, C− gave two families heterozygous to one homozygous susceptible. These two classes evidently overlap, and to include A+ in the homozygous resistant category, and C− in the heterozygous category, is not strictly valid. However if we assume that modifying genes are responsible for the relatively small overflow from the major gene categories we appear not to deviate far from the truth. Even with this error the F_2 distributions did not significantly deviate from the 1:2:1 ratio.

Backcrosses

Selected F_1 phenotypes in classes B and C− were backcrossed to susceptible and to resistant lines (Table 4). The progenies evidently conformed in general with the expectation of 1:1 ratios. Seven families deriving from susceptible parents were confined to the B, C− and C+ classes, except for a small number in the O to A− classes which are unexplained. As between the supposedly heterozygous category (B and C−) and the homozygous susceptibles, C+, there was no significant deviation from the 1:1 ratio, either in the separate groups or in the totals. Nevertheless the excess of B and C− over C+ suggests that errors in classification may have occurred and that some C− plants may have been in fact the homozygous susceptible genotype. We have already noted that in tests of the inbred homozygous susceptible, AFRO 54300, a small proportion of plants were classed C− and B.

Twelve families from resistant recurrent parents again conformed with the expected ratio, when classes O to A+ were combined as resistant and B to C+ as heterozygous. Deviations of the resulting totals (51·3–48·7%) from 1:1 were not significant.

DISCUSSION

From the results described, we conclude that a major gene gave primary control of resistance and susceptibility to the streak virus. Neither allelomorph was fully dominant; the heterozygote reacted to infection in a manner generally intermediate between the homozygous parents, but with a nearer approach to the susceptible. In field-grown plants the most severe effects are largely confined to the homozygous susceptible genotype; and indeed the heterozygote has been recommended as affording useful field control in South Africa (Gorter, 1959). However, there are evident advantages in breeding lines pure for the major gene for resistance.

After four generations of selfing, the most resistant selections from Peruvian × Hickory crosses from South Africa gave evidence that the major gene for resistance was present in homozygous form. It is true that the distributions in these families extended over the range of classes O, A− −, A− and rarely A+; but all these are effectively immune, and subdivision has no practical significance. The evidence suggests that modifying factors determine the small differences between these classes, but does not identify these factors. However, we have considered two possible non-

inherited factors: the temperature during the incubation of the virus in the plant, and efficiency in the individual vectors used. Neither can be responsible for heterogeneity in many replications in testing families of identical or similar origin through all seasons of the year. More probably, the modifications observed are due to minor genes.

In the inbred lines heterogeneity tends to be overemphasized by our adopting too fine a classification, particularly between grades of A, and it appears reasonable to class all A's with O in a resistant category. However, O parents gave families confined almost entirely to the resistant category, whereas some families from A− − and A− parents extended in parts into higher classes—results explicable by variations in the background of modifying genes.

The same general conditions appear to apply between lines carrying the two allelomorphs. Here, however, there was the possibility that further modifying genes might have been introduced in the susceptible parents, accounting for some diversity observed in the distributions. Despite this, the data do not conflict with the hypothesis of primary control by a major gene.

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EXPLANATION OF PLATE

Photographs of parts of leaves showing the typical range of symptoms in streak-diseased maize plants. Classification has been based on these symptoms. Magnification, *c.* × 1·5.

Fig. 1. Class A−. Typical also of class A− −. Symptoms consist of round chlorotic spots. (Confined on this leaf to the upper left-hand part of the leaf. The black markings are due to other causes.).

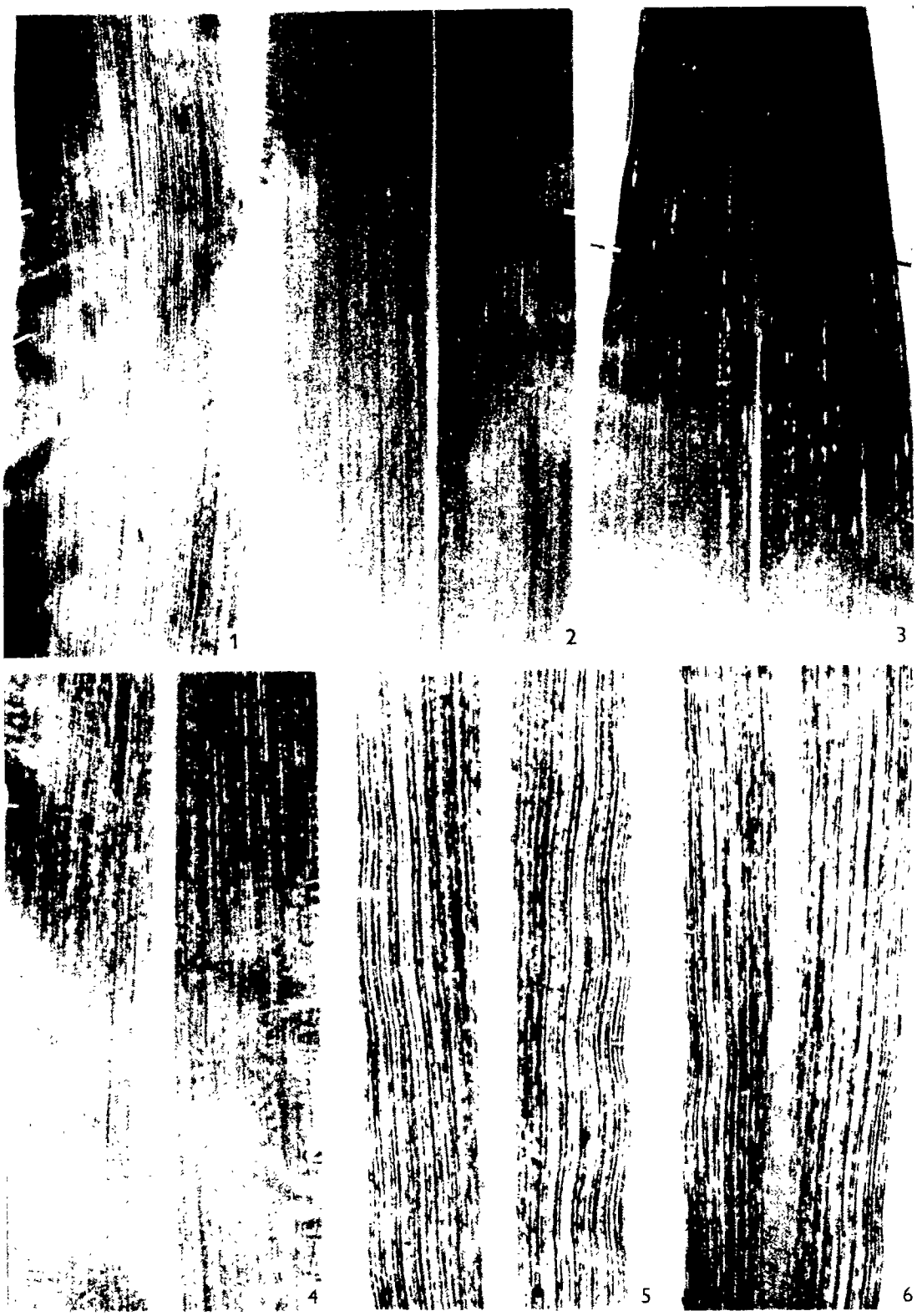
Fig. 2. Class A+. Spots as on fig. 1, but some tending to be elongated.

Fig. 3. Class B−.

Fig. 4. Class B+.

Fig. 5. Class C−.

Fig. 6. Class C+.



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